

Workshop Week 5 - Calculus 1 M
Winter 2024

Exercise 1. Find the slope of the tangent line at $(1, 1)$ to the curve given by the following equation:

$$e^{2x+3y} = e^5 x^2 - \ln(xy^3)$$

Exercise 2. Use the limits

$$\lim_{x \rightarrow 0} [1 + x]^{1/x} = e \text{ and } \lim_{t \rightarrow \pm\infty} [1 + \frac{1}{t}]^t = e$$

to prove a , b and c :

$$a) \lim_{x \rightarrow \infty} \left[\frac{x+8}{x} \right]^{x/3} = e^{8/3} \quad b) \lim_{x \rightarrow 3} [x^2 - 8]^{4/(x-3)} = e^{24} \quad c) \lim_{x \rightarrow \infty} \left[\frac{2^x - 3}{2^x + 3} \right]^{5^x} = 0$$

Exercise 3. We define the **hiperbolic sine** and **hiperbolic cosine functions**:

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \text{ and } \cosh(x) = \frac{e^x + e^{-x}}{2}$$

a) Notice that **hiperbolic sine** and **hiperbolic cosine** are differentiable functions on $\mathbb{R}$ and prove that $\sinh'(x) = \cosh(x)$ and $\cosh'(x) = \sinh(x) \forall x$.

b) Compute the domain of the function $f(x) = \frac{5}{(2 - \sinh(x))}$.

c) Find $f'(x)$ for all $x \in \text{Dom}(f)$. Is there any c such that $f'(c) = 2$? Justify carefully.
Hint: Use the Intermediate Value Theorem.

d) Find the equation of the tangent line of the inverse of the function above, f^{-1} , at $(c, f^{-1}(c))$ for $c = f(\ln 3)$.